ELSEVIER

Microelectronics Journal 37 (2006) 290–294

Microelectronics Journal

www.elsevier.com/locate/mejo

Modeling and gradient pattern analysis of irregular SFM structures of porous silicon

M.P.M.A. Baroni[a,*], R.R. Rosa[a], A. Ferreira da Silva[b], I. Pepe[b], L.S. Roman[c], F.M. Ramos[a], R. Ahuja[d], C. Persson[e], E. Veje[b]

[a]Núcleo para Simulação e Análise de Sistemas Complexos, Laboratório Associado de Computação e Matemática Aplicada, Instituto Nacional de Pesquisas Espaciais (INPE), 12201-970 São José dos Campos-SP, Brazil

[b]Grupo de Fisica Basica e Aplicada em Materiais Semicondutores, Laboratorio de Propriedades Opticas, Instituto de Fisica, Universidade Federal da Bahia, Campus Ondina, 40170-290 Salvador-BA, Brazil

[c]Grupo de Dispositivos Nanoestruturados, Departamento de Física, Universidade Federal do Paraná, 81531-990 Curitiba-PR, Brazil

[d]Department of Physics, Condensed Matter Theory Group, Uppsala University, SE-751 Uppsala, Sweden

[e]Applied Materials Physics, Department of Materials Science and Engineering, Royal Institute of Technology, SE-100 44 Stockholm, Sweden

Received 24 March 2005; received in revised form 16 May 2005; accepted 16 May 2005
Available online 15 August 2005

Abstract

Technological applications in opto-electronic devices have increased the interest in characterizing porous silicon structure patterns. Due to its physical properties, solutions from KPZ 2D are adopted to simulate the structure of porous material interface whose spatial characteristics are equivalent to those found in porous silicon samples. The analysis of the simulated and real Scanning Force Microscopy (SFM) surfaces was done using the Gradient Pattern Analysis (GPA). We found that the KPZ 2D model presented asymmetry levels compatible with the irregular surfaces observed by means of SFM images of π-Si.

Keywords: Porous silicon; KPZ equation; Gradient pattern analysis; Nanostructures

1. Introduction

Usually, the porous silicon samples are produced by anodic etching of crystalline silicon (c-Si) wafers in hydrofluoric (HF) acid solution. As reported by many authors (e.g.[1,2]) one of the main problem in the theory of porous silicon (π-Si) sample is that there is no satisfactory explanation on the possible correlation between their photoluminescense (PL) and their structural properties due to the different formation parameters (doping level, HF concentration and current density).

Recently, Ferreira da Silva et al. [3] have performed a gradient pattern analysis of a canonical sample set (CSS) of scanning force microscopy (SFM) images of π-Si. They applied the Gradient Pattern Analysis (GPA) [3,7] to images of three typical π-Si samples distinguished by different absorption energy levels and aspect ratios (low, intermediate and high roughness). The GPA is an innovative technique, which characterizes the formation and evolution of extended patterns based on the spatio-temporal correlations between large and small amplitude fluctuations of the structure represented as a gradient field [5,6,10] (see Appendix).

It is a known fact that higher porosity silicon exhibits a shift of the spectral position of the visible band to higher threshold energy, from 1.2 to 2.2 eV [3] (see Fig. 1). Asymmetric fragmentation parameter (the so-called first gradient moment $g_1^a(L)$) is highly sensitive to quantify asymmetric fine structures in $L \times L$ extended patterns. Therefore, an energy/structure correlation classification of the canonical π-Si samples, of the same size, by applying asymmetric fragmentation was used to characterize silicon porosity quantitatively. They showed that, for the canonical sample set, the only parameter showing a direct relationship of the structural asymmetry with PL energy is the first gradient moment (g_1^a). Taking into account this result one could interpret that global porosity is defined not only in

* Corresponding author.
E-mail address: mariana@lac.inpe.br (M.P.M.A. Baroni).

1879-2391/$ - see front matter © 2005 Elsevier Ltd. All rights reserved.
doi:10.1016/j.mejo.2005.05.029

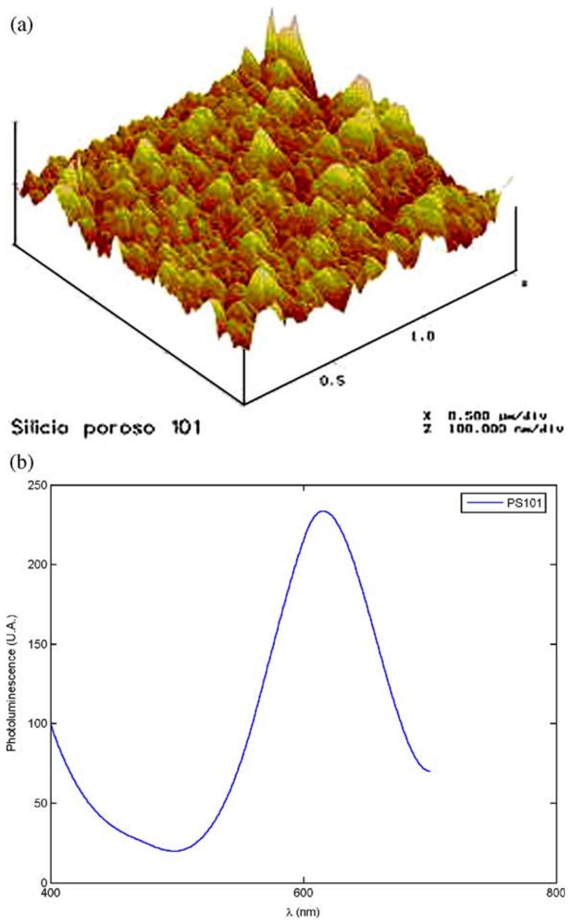

Fig. 1. SFM structural pattern (512×512) of a π-Si sample (up) and its photoluminescence measurement (down).

terms of the amount of roughness, but also in terms of the structural complexity of the roughness, mainly that described by means of the shape and size of asymmetric main structures as walls and fine structures as slots.

Recently, characterization of the universality class of π-Si has been performed taking into account a preferential direction of growth described for a Kardar-Parisi-Zhang (KPZ) model [11,12]. A comparative analysis and classification of both KPZ 2D and SFM π-Si structural patterns is considered discussing the importance of the modeling for the area of irregular nanostructured porous materials with prominence for the application in opto-electronic devices.

This analysis has been adapted in order to operate in an OpenGL flyby environment (the *StrFB code*), whose application gives the numerical characterization of the structure during the flyby in real time [10]. This application permits us to characterize, more accurately, the following structural parameters: aspect ratio (Γ) and the asymmetry gradient moment (g_1^a).

2. Modeling and numerical results

The structure formation models can be discrete or continuous [8], where the discrete models demand to simulate universal automata aspects of the problem. The continuous models generally use differential equations, with stochastic character features, where the main formation process obeys the general physical aspects of the problem. The continuous model used in this work is the KPZ equation.

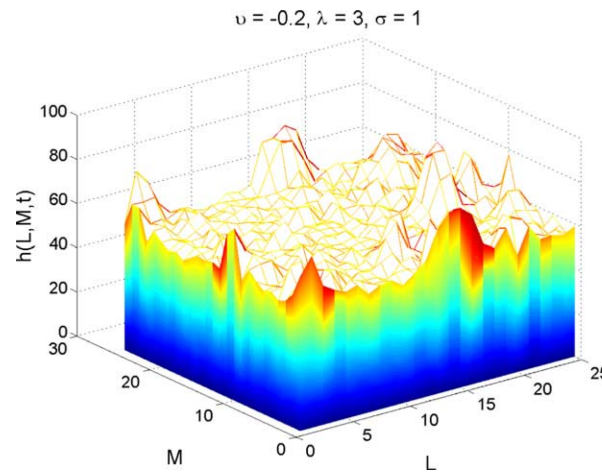

Fig. 2. An example of simulation by KPZ 2D model.

The KPZ 1D equation is given by the following expression [12]:

$$\frac{\partial h(x,t)}{\partial t} = \upsilon\nabla^2 h + \frac{\lambda}{2}(\nabla h)^2 + \eta(x,t)$$

where $h(x, t)$ is the height (position x, time t), υ is parameter of surface tension, λ is the parameter of lateral growth and $h(x,t)$ is the noise term. A result from KPZ 2D simulation, solved by means of the FTCS (forward time centered space) method [13], is shown in Fig. 2 where, the spatio-temporal scales ($L \times L$, with 0.1 µm per pixel) are compatible with the real π-Si scales.

This model presented results compatible with the irregular surfaces observed in the SFM samples of PS when their constitutive parameters (surface tension, noise term and lateral growth) are in the following ranges: $-0.3 \leq \upsilon \leq -0.1$; $-1 \leq \sigma \leq -0.5$ and λ without restriction. Table 1 and Fig. 3 present the best KPZ results.

3. Experimental results

In this paper, we have used three different typical-structure π-Si samples: Sample A, characterized by a low

Table 1
Morphological aspects for the best surfaces generated by the KPZ 2D model

KPZ 2D	Γ	g_1^a
A	0.5168	1.9728
B	0.6128	1.9744
C	0.8096	1.9728

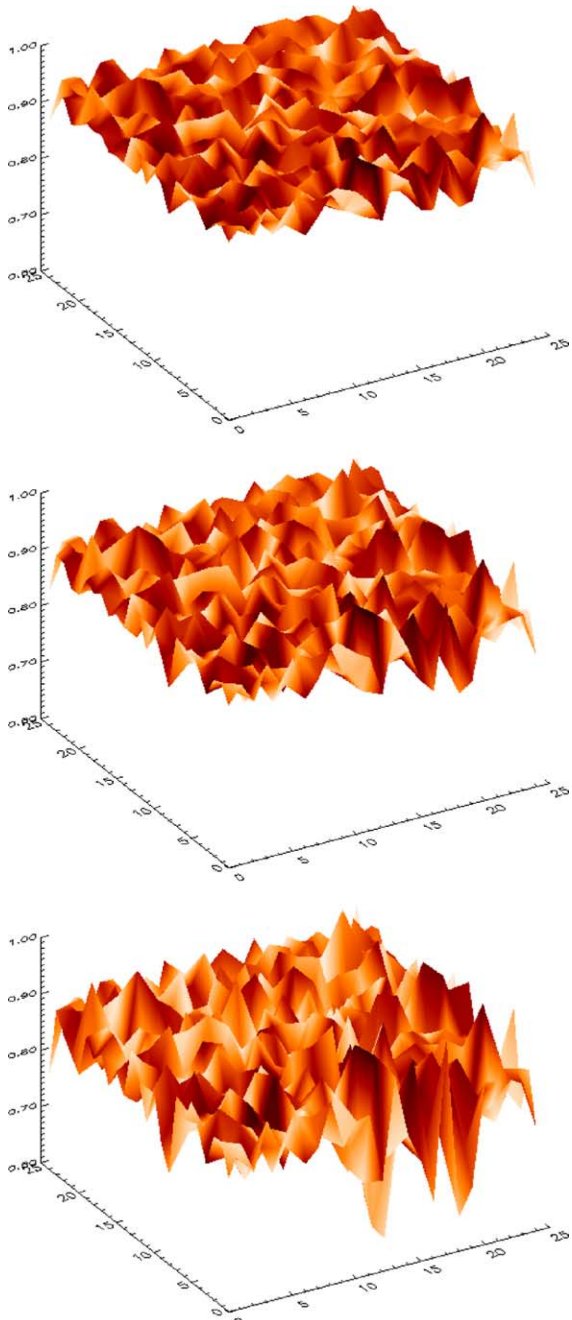

Fig. 3. The 25×25 grid (x,y) surfaces generated by the KPZ 2D model where Z is the normalized roughness amplitude best surfaces generated by the KPZ 2D model.

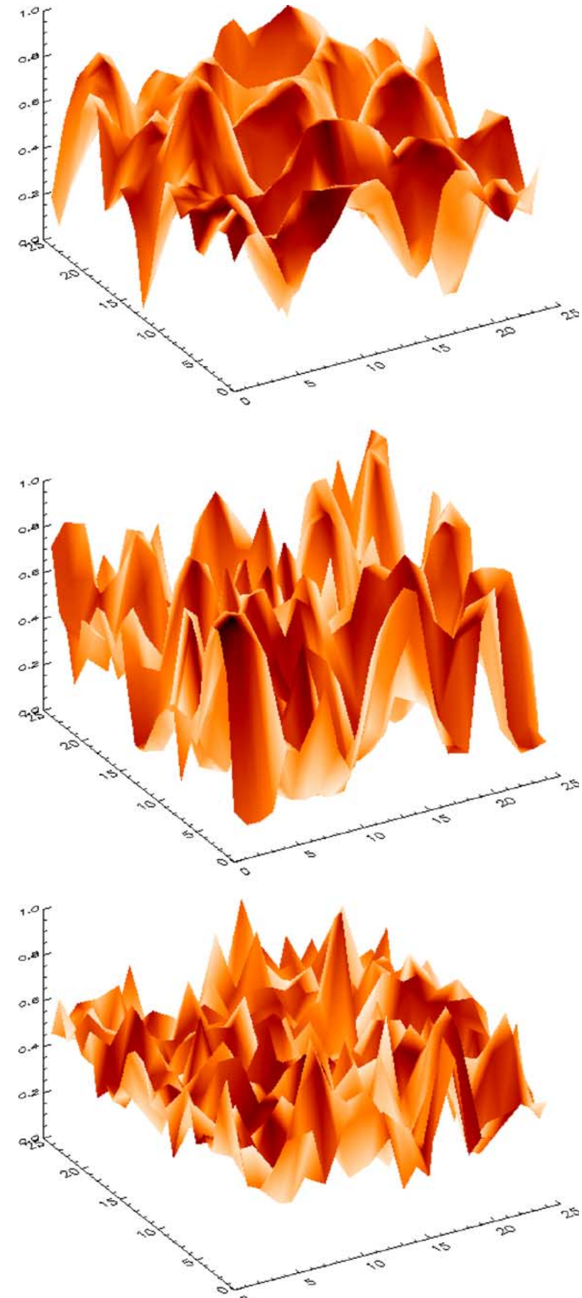

Fig. 4. SFM structural pattern of three PSi sample: (up) Sample A; (middle) Sample B and (bottom) Sample C.

roughness, Sample B of intermediate roughness and Sample C of high roughness (Fig. 4).

The porous silicon samples were produced by anodic etching of crystalline silicon wafers in hydrofluoric (HF) acid solution. The surface modification procedures have directed towards simproving the material optical properties. These samples were analyzed by Scanning Force Microscopy (SFM) to obtain the morphology of the porosity. The SFM measurements were performed using the Digital Instruments Nanoscope III running in tapping mode. In this imaging mode, the SFM tip-cantilever oscillated sinusoidally at high frequencies ($\sim$300 KHz) amplitude (10–100 nm), so that the tip contacted the surface once during each period. The data were taken for air at room temperature environment. Scans were made over areas from 500×500 nm till 20×20 μm with a resolution of 512×512 pixels and with the scan rate of 1–2 Hz.

The measured parameters for each sample are shown in Table 2, where J is current density and τ is the etching time. These are typical values used to obtain the π-Si by anodic

Table 2
The measured parameters for each sample

Sample	J (mA/cm2)	τ (min.)	$\bar{E}$ (eV)	Γ	g_1^a (25)
A	6.2	77	1.83	0.5760	1.9772
B	25	90	1.73	0.5904	1.9684
C	10	45	1.87	0.7392	1.9714

etching of c-Si wafers in HF acid solution [9]. The $\bar{E}$ is the absorption energy and Γ˙ is the degree of roughness determined from the calculation of the direct aspect ratio. This aspect ratio is defined as the ratio between the system dimension and the average of the least dimensions of characteristic scale (local cells).

4. Concluding remarks

This paper showed simulations and calculated the aspect ratio and asymmetry level for the KPZ 2D model compatible with the surfaces observed in the π-Si samples ($\Gamma = 0.64 \pm 0.09$, $g_1^a = 1.97 \pm 0.01$).

Based on this validation analysis the KPZ 2D surfaces are characterized by the following values: ($\Gamma = 0.65 \pm 0.15$, $g_1^a = 1.97 \pm 0.01$) where the input parameters for the KPZ numerical solutions are $-0.3 \leq \upsilon \leq -0.1$; $-1 \leq \sigma \leq -0.5$ and λ without restriction. More samples must be analyzed in order to study the robustness of this new characterization.

In summary, we have shown that the gradient pattern analysis technique in a flyby environment is a reliable method to investigate, qualitatively and quantitatively, the morphology of π-Si active porosity and we stress to discuss the importance of this approach into the field of nanofabrication.

Acknowledgements

The authors acknowledges financial support from FAPESP, CNPq, CNPq/NanoSemiMat under grant no. 550.015/01-9, the Swedish Foundation for International Cooperation in Research and Higher Education (STINT) and Swedish research Council (VR). M.P.M.A.B acknowledges Eronides F. da Silva Jr. for valuable suggestions and Nandamudi L. Vijaykumar for valuable English corrections.

Appendix: The gradient pattern analysis

In view of GPA formalism, the spatial structure fluctuation of a global pattern given by the matrix M(x,y), can be characterized by its gradient vector field G = ∇[M(x, y)]. The local spatial fluctuation, between each pair of pixels, of the global pattern is characterized by its gradient vector at corresponding mesh-points in the two-dimensional space. In this representation, the relative values between pixels are relevant, rather than the pixels' absolute values. Note that, in a gradient field such relative values, can be characterized by each local vector norm and its orientation. Thus, according to Rosa et al. [6], a given matricial scalar field can be represented as a composition of four gradient moments: g_1, is the integral representation of the vector distribution; g_2, is the integral representation of the norms; g_3, is the integral representation of the phases; and g_4, is the complex representation of the gradient pattern.

Considering the sets of local norms and phases as discrete compact groups, spatially distributed in a lattice, the gradient moments have the basic property of being, at least, rotational invariant. As we are interested in nonlinear extended structures we used a computational operator to estimate the gradient moment g_1 based on the asymmetries among the vectors of the gradient field of the scalar fluctuations. A global gradient asymmetry measurement, can be performed by means of the asymmetric amplitude fragmentation (AAF) operator [3]. This computational operator measures the symmetry breaking of a given dynamical pattern and has been used in several applications [3–6,10].

From ∇(M) the symmetric pairs of vectors (i.e. the pairs of vectors that have the same modulus but opposite directions) are removed, obtaining the asymmetric field of vectors ∇_A(M). The measurement of asymmetric spatial fragmentation g_1^a is defined as:

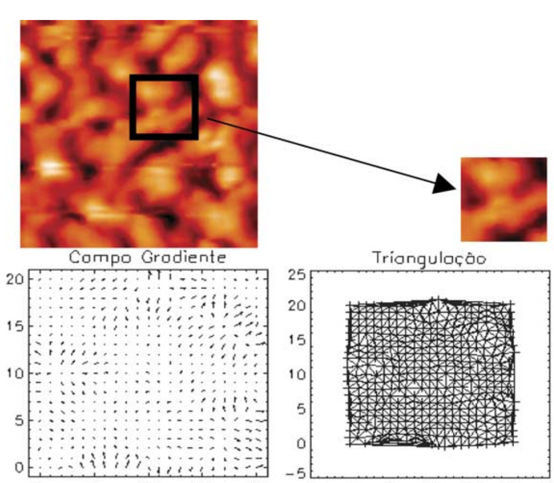

Fig. A1. GPA application result on a PS substructure image. (a) Gradient field; (b) Triangulation.

$$g_1^{\mathrm{a}} \equiv (C - V_{\mathrm{A}})/V_{\mathrm{A}} | C \geq V_{\mathrm{A}} > 0$$

where V_{A} is the number of asymmetric vectors and C is the number of correlation bars generated by a Delaunay triangulation having the middle point of the asymmetric vectors as vertices [6]. The Delaunay triangulation $T_{\mathrm{D}}(C, V_{\mathrm{A}})$ is a fractional field with dimension less than two—the lattice dimension. When there is no asymmetric correlation in the pattern, the total number of asymmetric vectors is zero, and then, g_1^{a} is null. Otherwise, this parameter quantifies the level of asymmetric fluctuations. In this report we are not interested in measurements of the higher order gradient moments. Several calculations on random patterns have shown that g_1 gradient moment is much more sensitive and precise in characterizing asymmetric structures than the correlation length measures [3,4]. When there is no asymmetric correlation in the pattern, the total number of asymmetric vectors is zero, and then, by definition g_1^{a} is null. For a random and totally disordered pattern, g_1^{a} has the highest value, for a totally ordered and locally symmetric pattern, g_1^{a} is equal to zero and for a complex pattern composed by locally asymmetric structures g_1^{a} is nonzero defining different classes of irregular patterns.

Fig. A1 shows an example on application of the GPA in a sub-image of porous silicon.

References

[1] L.T. Canham, *Appl. Phys. Lett.* 57 (1990) 1046.
[2] D.J. Lockwood, Solid State Commun. 92 (1994) 101.
[3] A. Ferreira da Silva, R.R. Rosa, L.S. Roman, E. Veje, I. Pepe, Solid State Commun. 113 (2000) 703.
[4] R.R. Rosa, A.S. Sharma, J. Valdivia, *Int. J. Mod. Phys. C* 10 (1999) 147.
[5] F.M. Ramos, R.R. Rosa, C. Rodrigues Neto, Physica A 283 (2000) 171.
[6] R.R. Rosa, M.R. Campos, F.M. Ramos, S. Fujiwara, T. Sato, Braz. J. Phys. 33 (2003) 605.
[7] A. Ferreira da Silva, R.R. Rosa, P.W.A. Machado, F.M. Ramos, C. Rodrigues Neto, L.S. Roman, E. Veje, Physica A 283 (2000) 223.
[8] A.L. Barabasi, H.E. Stanley, Fractal concepts in surface growth, Cambridge University Press, Oxford, NY, 1994.
[9] O.K. Andersen, E. Veje, Phys. Rev. B 53 (1996) 15643.
[10] R.R. Rosa, A. Ferreira da Silva, R.C. Brito, L.S. Roman, M.P.M.A. Baroni, F.M. Ramos, R. Ahuja, C. Persson, Phys. Stat. Solid (c) 1 (2004) 277.
[11] M.E.R. Dotto, M.U. Kleinke, Phys. Rev. B 65 (2002) 245–323.
[12] M. Kardar, G. Parisi, Yi-C Zhang, Phys. Rev. Lett. 56 (1986) 889.
[13] M.P.M.A. Baroni, MS Thesis, INPE (2005).